DATA ANALYSIS PROJECT · DIGITERA TRAINING

# PharmaPulse Analytics

## Sales, Loss & Inventory Intelligence for Pharmacy Chains

End to end: Python · SQL Server · Power BI · DAX

[Your name] · Junior Data Analyst

# The business problem

A pharmacy chain with 12 stores wants to know where profit leaks, which discounts pay off, and where stock is at risk.

**Raw data**
3 sales files + 8 tables

**Python**
Clean and merge

**SQL Server**
Explore and validate

**Power BI**
Model, DAX, dashboard

**Insights**
Decisions

# Python: cleaning the data

```
# two date formats in the raw files
is_slash = d.str.match(r"^\d{2}/\d{2}/\d{4}$")
parsed[is_slash]  = to_datetime(d[is_slash],  format="%d/%m/%Y")
parsed[~is_slash] = to_datetime(d[~is_slash], format="%Y-%m-%d")
# flag the sales that lose money
sales["is_loss"] = (sales["profit"] < 0).astype(int)
```

## What the script handles

- Merges 3 sales files
- Removes 50 duplicate rows and 42 duplicate customers
- Extracts promo codes from free-text notes
- Adds profit, margin, is_loss, discount_band

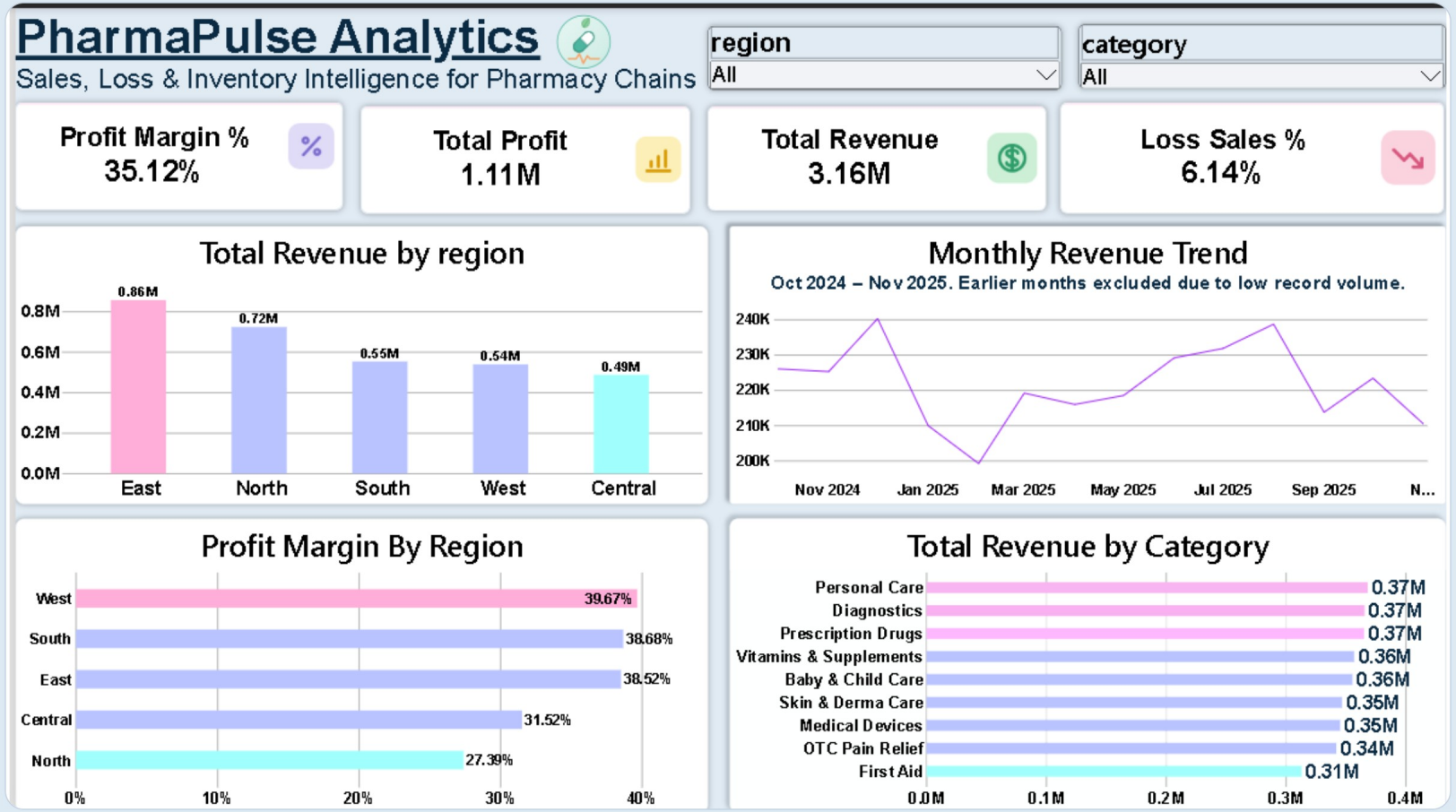
# SQL: do bigger discounts hurt margin?

```
SELECT discount_band,
  ROUND(100.0*SUM(profit)/SUM(total_amount),2) AS margin_pct,
  ROUND(100.0*SUM(is_loss)/COUNT(*),2) AS loss_rate_pct
FROM dbo.Sales_cleaned
GROUP BY discount_band;
```

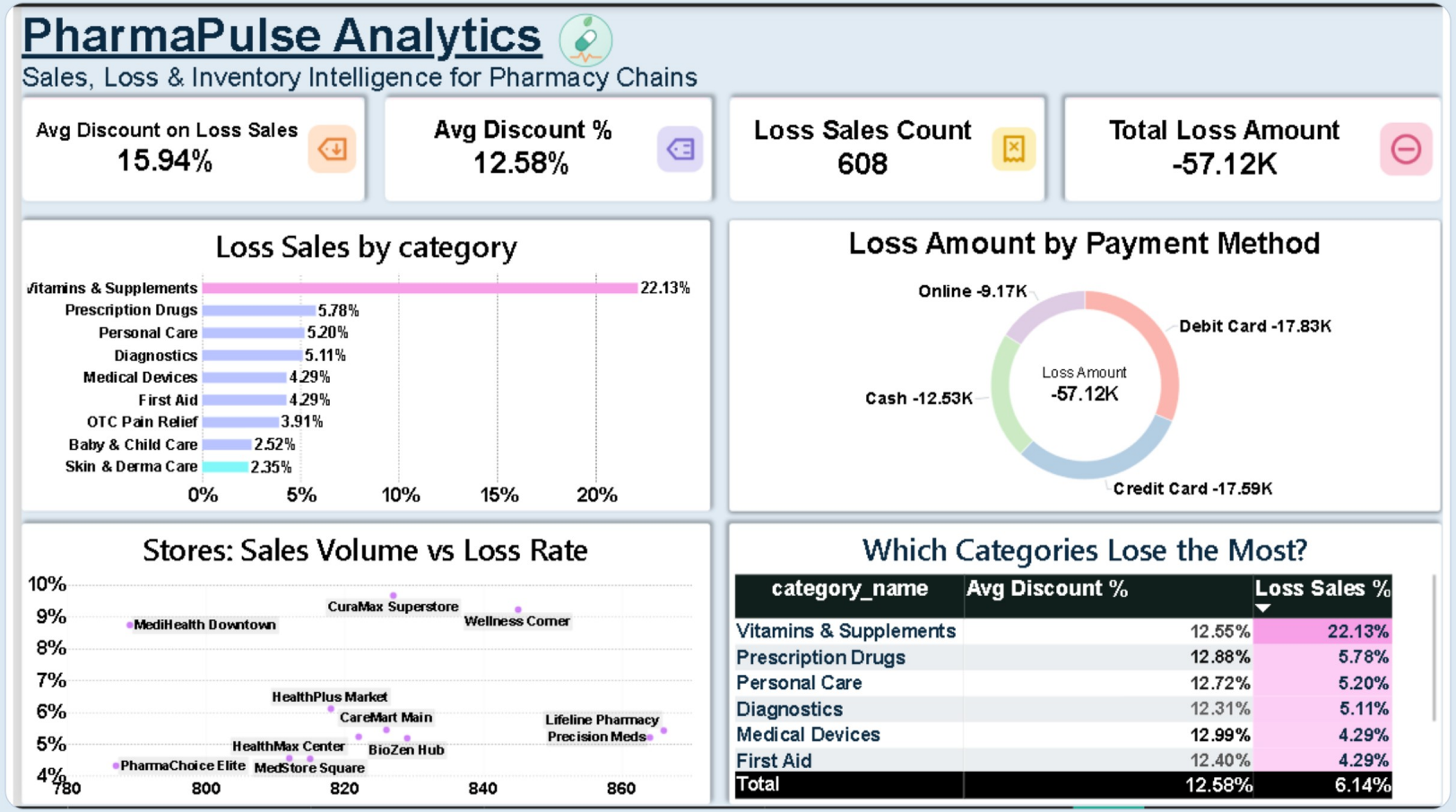
| Discount | Margin | Loss rate |
|---|---|---|
| 0-5% | 40.9% | 4.8% |
| 5-10% | 38.7% | 5.2% |
| 10-15% | 36.5% | 5.2% |
| 15%+ | 21.3% | 9.2% |

21 queries in total: JOINs, CASE WHEN, HAVING, CTEs, LAG and RANK.

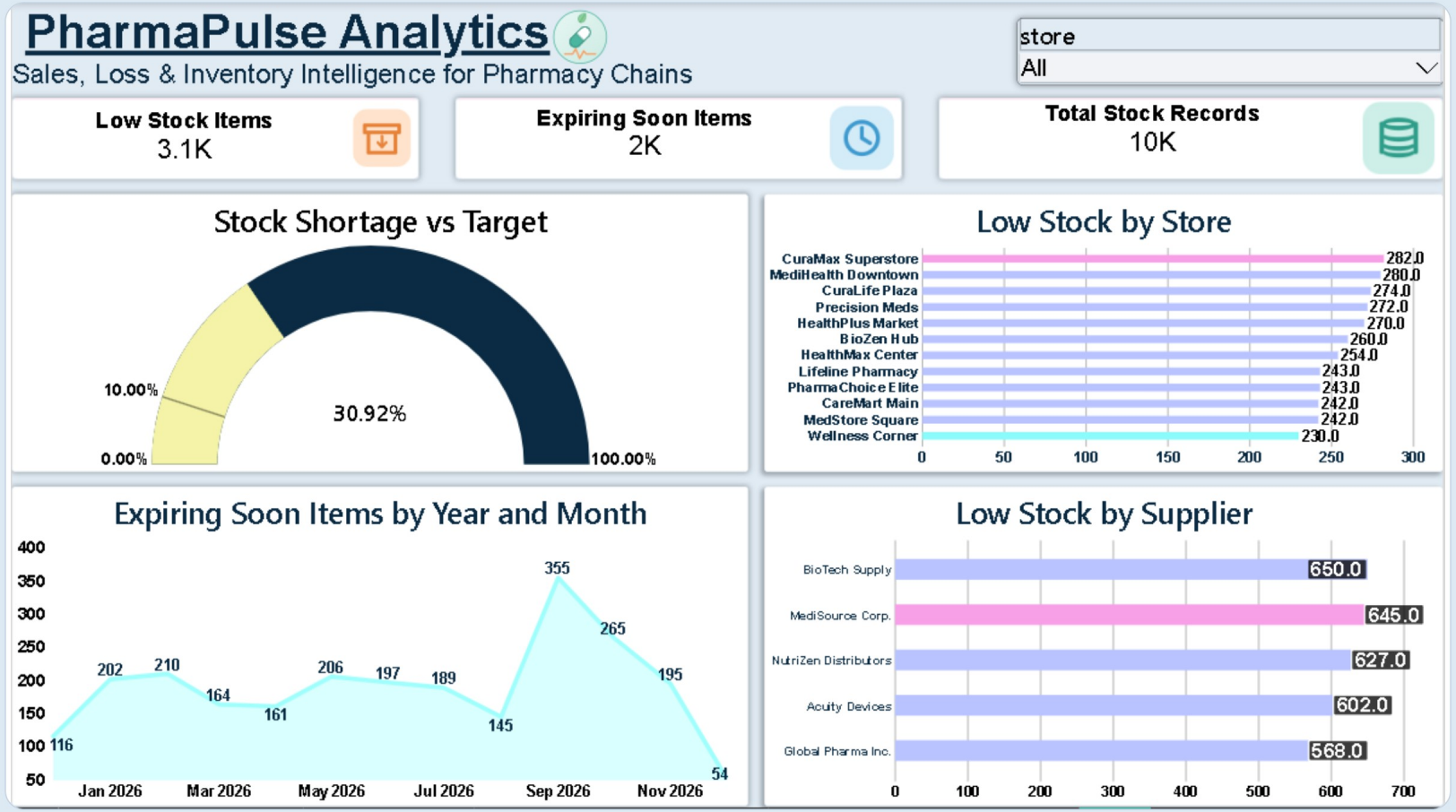
# Overview: East earns the most, West has the best margin



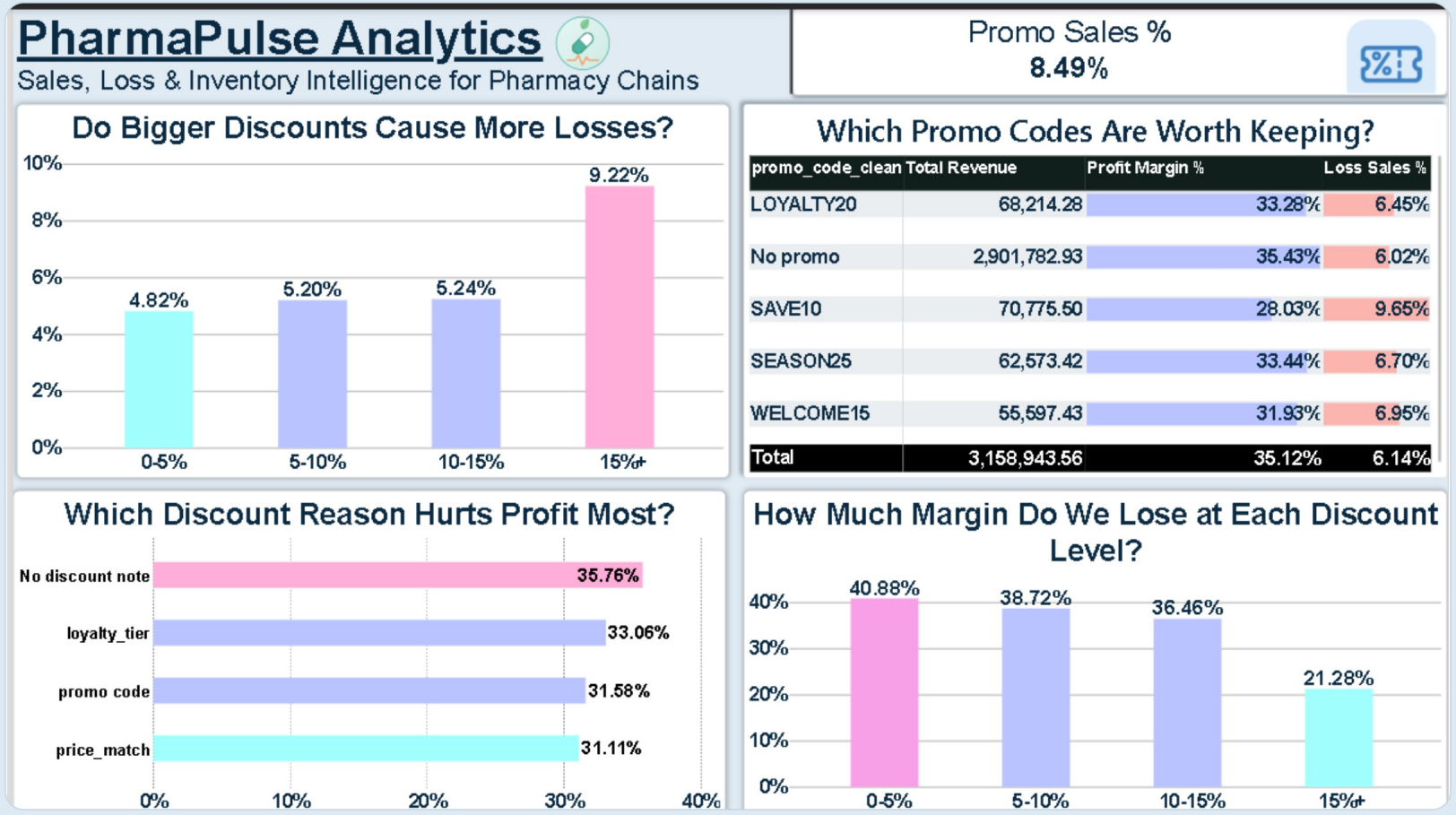
# Loss: Vitamins lose 22% of their sales



# Inventory: 31% of stock records are below reorder level



# Discounts: above 15%, the margin is cut in half



# What the data says

**21.3%**

margin on discounts above 15%, against 40.9% for 0-5%. Loss rate doubles to 9.2%.

**22.1%**

loss rate in Vitamins. 112 of its 543 products are priced below cost.

**28.0%**

margin on promo code SAVE10, the weakest code, with a 9.7% loss rate.

**Recommendations:** cap discounts near 15%, reprice the Vitamins products sold below cost, review SAVE10.

# Thank you

Python · SQL Server · Power BI · DAX · Excel

Code, queries and dashboard: github.com/[username]/PharmaPulse-Analytics

[Your name] · Junior Data Analyst · [LinkedIn link]